NIKHIL MURUGAN D P

Software Developer | AI/ML | Cloud Computing | Full-Stack Development

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PROFESSIONAL SUMMARY

MCA student at Vellore Institute of Technology with a strong interest in software development, artificial intelligence, cloud computing, and emerging technologies. Skilled in multiple programming languages with hands-on experience developing full-stack applications, AI-driven systems, cloud-integrated solutions, and research-oriented software. Experienced in building an intelligent EV mobility application, a multi-agent research intelligence framework using LLMs, RAG and Agentic AI, and a multi-tenant SaaS platform using modern web technologies and databases. Strong foundation in software engineering, machine learning, database management, application development, problem-solving, and system design, with a strong interest in applying emerging technologies to develop practical, scalable, and user-focused software solutions.

EDUCATION

Master of Computer Applications **2025–2027**
Vellore Institute of Technology, Vellore

Bachelor of Computer Applications **2022–2025**
Vellore Institute of Technology, Vellore

SKILLS

Programming Languages: Python, Java, C, C++, PHP

Web Technologies: HTML, CSS, JavaScript, React.js, Angular, Node.js, REST APIs, MERN Stack Development

Software Engineering: Software Testing, Software Project Management, SDLC Concepts, Agile Fundamentals

Frameworks & Tools: Flask, Git, GitHub, VS Code, MongoDB Compass, Postman, XAMPP, AWS, Supabase, Google Colab

Database Management: SQL, MySQL, MongoDB, PostgreSQL

AI & ML: Machine Learning, Deep Learning, Predictive Analytics, LLMs, RAG, Agentic AI, Generative AI

Cybersecurity, Cloud & DevOps: Cybersecurity Fundamentals, Cloud Computing, DevOps Basics

Core Technical Skills: Problem Solving, Data Structures & Algorithms, System Design, Computer Networks, Distributed Systems

PROJECT EXPERIENCE

ElectraNav — Electric Vehicle Predictive Navigation System with Intelligent Charging Station Discovery and Energy Optimization

- Developed an intelligent EV mobility application using Dart and Flutter, supported by a Python Flask backend, to assist users with route planning, charging-station discovery, battery analysis, energy prediction, and trip decision support.
- Implemented AI/ML-based energy prediction and battery intelligence to estimate energy consumption, battery state, remaining driving range, charging requirements, and battery utilization for planned journeys.
- Integrated intelligent route analysis, charging feasibility evaluation, trip analytics, and Digital Twin-based battery simulation to support comparative and energy-aware EV trip planning.

ResearchPilotAI — A Multi-Agent Research Intelligence Framework for Autonomous Scientific Discovery and Knowledge Synthesis

- Designed and developed ResearchPilotAI as a complete AI-based research intelligence system using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Agentic AI to support intelligent research workflows.
- Implemented multi-agent reasoning and task orchestration to perform knowledge discovery, technical document analysis, research synthesis, and generation of structured research-oriented outputs.
- Developed an intelligent knowledge-processing workflow capable of working with complex technical information and transforming research content into organized, contextual, and meaningful insights.

Nexora — Integrated Multi-Tenant SaaS Workflow and Organizational Intelligence System for Enterprise Startup Operations and Business Management

- Designed and developed a multi-tenant SaaS platform using React, Angular, Node.js, and MongoDB to provide a unified environment for organizational workflows, project management, team operations, CRM, and document management.
- Implemented modular application features including dashboards, projects, tasks, team members, CRM leads, documents, and Kanban-based workflow management with structured database operations.
- Developed a scalable software architecture focused on multi-tenant workflows, role-based organizational operations, business process management, and centralized data management.

CERTIFICATIONS & LEARNING ACHIEVEMENTS

AWS Job Roles in the Cloud — Foundational Cloud Computing, AWS Training & Certification (Oct 2024)

Microsoft Learn — Generative AI Foundations and Applied AI Learning Achievements

LINKS

LinkedIn LeetCode GitHub